Project: Worked examples
Title: Accumulator sizing (gas law)
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1. Scope and method

Calculates the pre-charge gas volume V0 of a hydraulic accumulator that delivers or absorbs a liquid volume ΔV between two working pressures, with the next listed size and the PSSR bar·litre product. It is not a pressure-vessel code check.

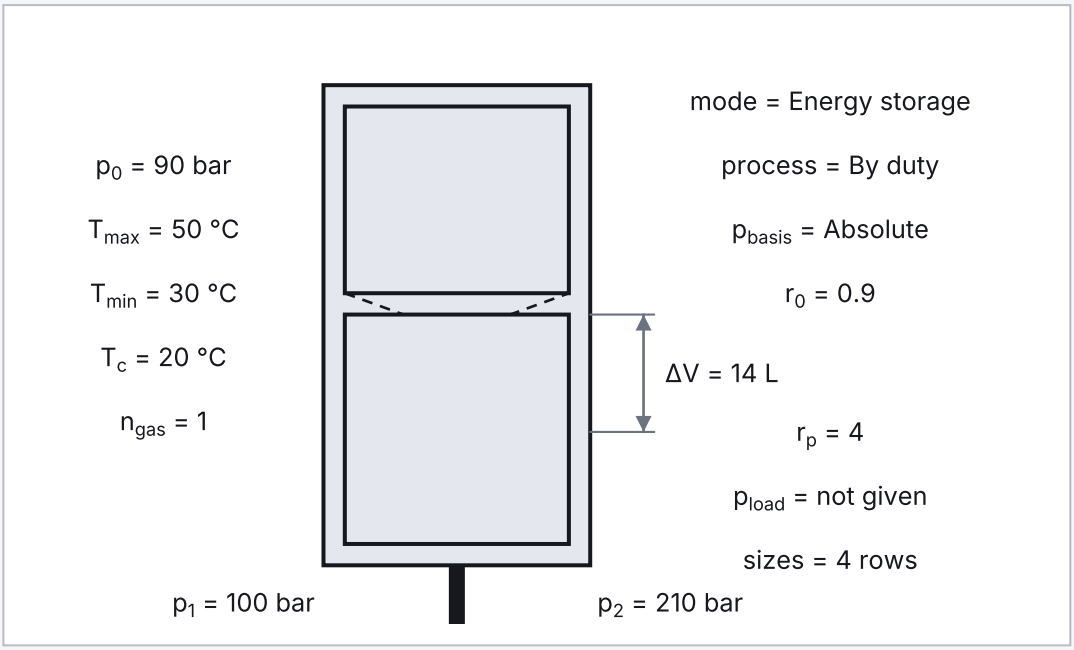


Figure 1 — A bladder accumulator in section: gas pre-charged to p_0 above the bladder, the liquid ΔV delivered between p_2 and p_1 through the port. Not to scale; the labels are the inputs as entered.

WHAT IT CALCULATES	RELATION
Absolute pressures	$p_a = p_{gauge} + p_{atm}$, or as entered in Absolute mode
Gas volume, one exponent both ways	$V0 = \Delta V / [(p0/p1)^{(1/n_{gas})} - (p0/p2)^{(1/n_{gas})}]$, $n_{gas} = 1$ isothermal to 1.4 adiabatic
Gas volume, slow charge then discharge at n_{gas}	$V0 = \Delta V \cdot (p2/p0) / [(p2/p1)^{(1/n_{gas})} - 1]$, from $V2 = V0 \cdot p0/p2$ and $V1 = V2 \cdot (p2/p1)^{(1/n_{gas})}$
Temperature and gas corrections	$V0 \times f_{rg} \times$ the cold-gas factor T_{max}/T_{min}, to the power $1/n_{gas}$ when one exponent holds both ways, in absolute temperatures
Pre-charge rules	$p0 \le r_0 \cdot p1$ at T_{max}, and $p2/p0(T_{min}) \le r_p$ with $p0(T_{min}) = p0 \cdot T_{min}/T_{max}$ in absolute temperatures
Pre-charge at the charging temperature	$p0(T_c) = p0(T_{max}) \cdot T_c/T_{max}$ in absolute temperatures
Next size	the smallest listed size not below V0
PSSR bar-litre product	PS·V of the vessel against 250 bar·L, an indication that needs PS and a volume

Basis: 250 bar·litre exception, indication only[1](§Schedule 1); Pressure equipment, not checked here[2]; Gas-charged accumulator sizing, $p \cdot V^n$ = const[3](Ch. on accumulators, 7th ed.).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
mode	Duty	Energy storage		assumption: Slow clamp-and-hold cycle: stored-energy duty.	any
process	Gas process (By duty: an emergency stroke is charged slowly to p_2 and discharged at n_{gas}, the other duties use one exponent both ways)	By duty		assumption: By duty: a stored-energy duty uses the one exponent n_gas both ways, isothermal here.	any
p_{basis}	Pressures entered as	Absolute		assumption: Pressures entered as absolute bar, as the source numbers are quoted.	any
dV	Liquid volume delivered, absorbed or needed for the emergency stroke	14	L	spec: Hydraulic specification HS-12: clamp stroke plus leakage make-up, 14 L per cycle	0.01 … 500 L
p_0	Gas pre-charge pressure p0, at the maximum operating temperature	90	bar a	spec: Hydraulic specification HS-12: pre-charge 90 bar at the maximum oil temperature	1 … 400 bar a
p_1	Lowest working pressure p1	100	bar a	spec: Hydraulic specification HS-12: lowest clamp pressure 100 bar	1 … 400 bar a
p_2	Highest working pressure p2	210	bar a	spec: Hydraulic specification HS-12: pump cut-out 210 bar	1 … 400 bar a
p_{atm}	Atmospheric pressure (gauge to absolute)	1.01325	bar a	assumption: Standard atmosphere, derived constant.	0.7 … 1.05 bar a
T_{max}	Maximum operating temperature (p0 applies here)	50	°C	spec: Hydraulic specification HS-12: oil temperature limit 50 °C	-20 … 120 °C

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
T_{min}	Minimum operating temperature (the gas is coldest and delivers least here)	30	°C	assumption: Oil temperature range assumed 30 to 50 °C in service (invented for this example); the gas is coldest at 30 °C.	-30 ... 100 °C
T_c	Charging temperature	20	°C	assumption: Charged in the workshop at 20 °C.	-20 ... 60 °C
n_{gas}	Gas exponent of the discharge, 1 isothermal (slow) to 1.4 adiabatic (fast), also the charge exponent when one exponent holds both ways	1		assumption: Isothermal: the clamp holds for minutes between cycles.	1 ... 1.4
f_{rg}	Real-gas factor on V0, the real over the ideal gas volume for this pressure window (1 = ideal gas, at least 1)	1		assumption: Ideal gas; the highest pressure is just above 200 bar, so the sheet warns.	1 ... 2.1
r_0	Pre-charge rule: p0 (at T_{max}) at most this fraction of p1, on absolute pressures	0.9		reference: record: accumulator pre-charge rules (bladder type), to be taken from the manufacturer's guide	0.5 ... 0.95
r_p	Pre-charge rule: p2/p0 (p0 at T_{min}) at most this ratio, on absolute pressures	4		reference: record: accumulator pre-charge rules (bladder type), to be taken from the manufacturer's guide	2 ... 10
sizes	Available accumulator sizes from the manufacturer's range (optional; blank = no size is selected)	4 row(s)		reference: record: illustrative size list invented for this example, to be replaced by the manufacturer's range	any
PS	Vessel maximum allowable pressure, bar gauge, from the nameplate or relief setting (optional; blank = no PSSR indication)	230	bar g	spec: Hydraulic specification HS-12: accumulator relief setting 230 bar gauge (vessel PS; illustrative, to be replaced by the nameplate)	10 ... 700 bar g

TABLE ROW	VALUES
sizes[1]	V = 10 L
sizes[2]	V = 20 L
sizes[3]	V = 32 L
sizes[4]	V = 50 L

3. Assumptions

A1. All gas-law pressures are absolute. ($p \cdot V^n$ = const on a fixed mass of gas.)
A2. A gauge entry is converted with the atmospheric pressure entered on the sheet. ($p \cdot V^n$ = const on a fixed mass of gas.)
A3. The exponent n_{gas} is yours, from 1 for a slow isothermal discharge to 1.4 for a fast adiabatic one. (Gas-law exponents, closed form.)
A4. With one exponent both ways, a larger n_{gas} gives a larger volume for a given ΔV. (Gas-law exponents, closed form.)
A5. A held accumulator is charged slowly and isothermally to p_2, then discharged at n_{gas}. (Gas-law sizing of a held accumulator: slow charge, fast discharge.)
A6. An emergency stroke is a held accumulator by default. (Gas-law sizing of a held accumulator: slow charge, fast discharge.)
A7. Sizing a held accumulator with one exponent understates the volume. (Gas-law sizing of a held accumulator: slow charge, fast discharge.)
A8. The one-exponent form suits a gas charged as fast as it is discharged. (Gas-law sizing of a held accumulator: slow charge, fast discharge.)
A9. The shock and pulsation duty always uses one exponent. (Gas-law sizing of a held accumulator: slow charge, fast discharge.)
A10. The gas is ideal unless a real-gas factor is entered. (Real-gas correction as a user factor.)
A11. The real-gas factor is at least 1, because real nitrogen needs more volume than ideal gas, never less. (Real-gas correction as a user factor.)
A12. Above 200 bar the sheet warns that nitrogen departs from the ideal-gas law. (Real-gas correction as a user factor.)
A13. The pre-charge p0 is defined at the maximum operating temperature T_{max}, and V0 is sized there. (Manufacturer sizing practice and the gas law at constant mass.)
A14. At T_{min} the same gas delivers only the fraction T_{min}/T_{max} of ΔV in absolute temperatures. (Manufacturer sizing practice and the gas law at constant mass.)
A15. A held accumulator multiplies V0 by T_{max}/T_{min}. (Manufacturer sizing practice and the gas law at constant mass.)
A16. One exponent n_{gas} both ways multiplies V0 by T_{max}/T_{min} to the power $1/n_{gas}$. (Manufacturer sizing practice and the gas law at constant mass.)
A17. The p2/p0 rule is checked with the pre-charge at T_{min}. (Manufacturer sizing practice and the gas law at constant mass.)
A18. With T_{min} equal to T_{max} the temperature correction is 1. (Manufacturer sizing practice and the gas law at constant mass.)
A19. The pre-charge rules are records for the accumulator type and make. (Rules are records, not built-in limits.)
A20. The rules are checked on absolute pressures, whichever basis the pressures are entered on. (Rules are records, not built-in limits.)
A21. The manufacturer gives the rule values and says whether its ratio is on gauge or absolute pressures. (Rules are records, not built-in limits.)
A22. The bar·litre product is PS times the vessel volume V. (Regulatory gating: no pressure-vessel code check.)
A23. PS is the vessel's maximum allowable pressure in bar gauge, never below the highest working

pressure p_2. (Regulatory gating: no pressure-vessel code check.)
A24. The vessel volume is the selected size unless you enter it. (Regulatory gating: no pressure-vessel code check.)
A25. Without PS and a volume the PSSR indication is not available. (Regulatory gating: no pressure-vessel code check.)
A26. The indication shows whether the 250 bar·L exception can apply, and it is not a compliance statement. (Regulatory gating: no pressure-vessel code check.)
A27. The PED, UK PER, PSSR examination and the vessel's design code stay with the manufacturer and the competent person. (Regulatory gating: no pressure-vessel code check.)
A28. mode (Duty) is taken as Energy storage. (Slow clamp-and-hold cycle: stored-energy duty.)
A29. process (Gas process (By duty: an emergency stroke is charged slowly to p_2 and discharged at n_{gas}, the other duties use one exponent both ways)) is taken as By duty. (By duty: a stored-energy duty uses the one exponent n_{gas} both ways, isothermal here.)
A30. p_{basis} (Pressures entered as) is taken as Absolute. (Pressures entered as absolute bar, as the source numbers are quoted.)
A31. p_{atm} (Atmospheric pressure (gauge to absolute)) is taken as 1.01325 bar a. (Standard atmosphere, derived constant.)
A32. T_{min} (Minimum operating temperature (the gas is coldest and delivers least here)) is taken as 30 °C. (Oil temperature range assumed 30 to 50 °C in service (invented for this example); the gas is coldest at 30 °C.)
A33. T_c (Charging temperature) is taken as 20 °C. (Charged in the workshop at 20 °C.)
A34. n_{gas} (Gas exponent of the discharge, 1 isothermal (slow) to 1.4 adiabatic (fast), also the charge exponent when one exponent holds both ways) is taken as 1. (Isothermal: the clamp holds for minutes between cycles.)
A35. f_{rg} (Real-gas factor on V0, the real over the ideal gas volume for this pressure window (1 = ideal gas, at least 1)) is taken as 1. (Ideal gas; the highest pressure is just above 200 bar, so the sheet warns.)
A36. Energy storage duty: ΔV is the liquid volume delivered between p_2 and p_1 over a cycle, with n_{gas} from 1 for a slow discharge to 1.4 for a fast one. (Gas-law sizing for a stored-energy duty.)

4. Working

[15] $pa_0 = p_0$
= 90 bar a
= 90 bar a
Pre-charge pressure, absolute

[16] $pa_1 = p_1$
= 100 bar a
= 100 bar a
Lowest working pressure, absolute

[17] $pa_2 = p_2$
= 210 bar a
= 210 bar a
Highest working pressure, absolute

[18] $p2_g = \max(pa_2 - p_{atm}, 0\ bar)$
= max(210 bar a − 1.0132 bar a, 0 bar)
= 208.99 bar g
Highest working pressure, gauge

[21] $V0_{ideal} = dV / [(pa_0/pa_1)^{(1/n_{gas})} - (pa_0/pa_2)^{(1/n_{gas})}]$ [4](const)
= 14 L / [(0.9) − (0.42857)]
= 29.697 L
Gas volume for the displacement, ideal gas

[22] $f_{Tabs} = (T_{max} / 1\ K) / (T_{min} / 1\ K)$
= 50 degC / 1 K / (30 degC / 1 K)
= 1.066

Absolute temperature ratio T_{max}/T_{min}

[23] $f_T = f_{Tabs}\text{^}(1 / n_{gas})$

= 1.066^(1 / 1)

= 1.066

Temperature scaling of the gas volume: the temperature ratio to the power 1/n (one exponent both ways)

[24] $V0 = V0_{ideal} * f_{rg} * f_T$

= 29.697 L × 1 × 1.066

= 31.656 L

Pre-charge gas volume V0 (accumulator size required)

[25] $T_{ratio} = (T_c / 1\ K) / (T_{max} / 1\ K)$

= 20 degC / 1 K / (50 degC / 1 K)

= 0.90716

Charging over maximum absolute temperature

[26] $p0 = pa_0 * T_{ratio}$

= 90 bar a × 0.90716

= 81.645 bar a

Absolute pre-charge pressure at the charging temperature

[27] $p0_{min} = pa_0 / f_{Tabs}$

= 90 bar a / 1.066

= 84.43 bar a

Absolute pre-charge pressure at the minimum operating temperature

[28] $p0_{set} = p0$

= 81.645 bar a

= 81.645 bar a

Pre-charge to set at the charging temperature (absolute)

[29] $p0_{max} = r_0 * pa_1$

= 0.9 × 100 bar a

= 90 bar a

Largest pre-charge the p0/p1 rule allows

[30] $u_{r0} = pa_0 / p0_{max}$

= 90 bar a / 90 bar a

= 1

Pre-charge against the p0/p1 rule

[31] $p_{ratio} = pa_2 / p0_{min}$

= 210 bar a / 84.43 bar a

= 2.4873

Pressure ratio p2/p0, with p0 at the minimum temperature

[32] $u_{rp} = p_{ratio} / r_p$

= 2.4873 / 4

= 0.62182

Pressure ratio p2/p0 (p0 at the minimum temperature) against the rule

[33] V_{sel} = smallest listed size not below V0

= V0 = 31.656 L, 4 sizes listed

= 32 L

Selected accumulator size

[34] $u_V = V0 / V_{sel}$

= 31.656 L / 32 L

= 0.98926

Required volume over the selected size

[35] $bar_{litres} = PS * V_{sel}$

= 230 bar g × 32 L

= 7360 bar·L

Pressure-volume product PS·V (PS as entered, V the selected size)

[36] BL_{pssr} = 250 bar∗L [5](1)
= 250 bar·L
PSSR Schedule 1 exception threshold (indication)

5. Results

SYMBOL	VALUE	DESCRIPTION
V0	31.66 L	Accumulator gas volume required
p0	81.64 bar a	Pre-charge, absolute, at the charging temperature
bar_{litres}	7360 bar·L	Pressure-volume product (bar·litres)
V_{sel}	32 L	Selected size
$p0_{set}$	81.64 bar a	Pre-charge to set at the charging temperature
$p0_{min}$	84.43 bar a	Pre-charge, absolute, at the minimum operating temperature
f_T	1.066	Gas volume scaling T_{max}/T_{min}
pssr	"7360 bar·L is at or above 250 bar·L: the exception does not apply; PSSR and PED / UK PER duties follow."	PSSR Schedule 1 250 bar·L exception (indication only)
process	"charged as fast as it is discharged (one exponent n_{gas} = 1 from the pre-charge to both working pressures)"	Gas process the volume is sized for
u_V	0.99	Selected size utilisation
u_{r0}	1.00	Pre-charge rule utilisation (p0/p1) (governing)
u_{rp}	0.62	Pressure ratio utilisation (p2/p0 at the minimum temperature)

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Pre-charge against the p0/p1 rule	90 bar a	90 bar a	1.00	[6](record)
PASS	Pressure ratio against the p2/p0 rule	2.4873	4	0.62	[6](record)

7. Applicability

WARNING records: tabulated values not taken from a reference record are r_0, r_p (Pre-charge rules); sizes (Accumulator size list). Enter each as one so its source, edition and licence travel with it. [record]
WARNING p_2: the highest working pressure is 210 bar absolute, above 200 bar, where nitrogen departs from the ideal gas. The volume shown is too small, so enter f_{rg} (up to 3) from a gas-property source. [realgas]

8. Verification

```
Template fp.accumulator@1.7.0 Content hash 7ea8865ce36556d1 Engine calc-engine/1.0.0
Known-answer cases passed 10/10 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```

9. References

1. Pressure Systems Safety Regulations 2000 (SI 2000/128).
2. Pressure Equipment Directive 2014/68/EU and UK PER 2016.
3. Esposito, Fluid Power with Applications.
4. p·Vⁿ =.
5. PSSR Schedule.
6. accumulator.rules.